

EEE308L

Electronic Circuits II Laboratory

Lab Report

Section:06

Group:04

Experiment :07

Name of the Experiment: *Study of Amplifier as RC Phase Shift Oscillator*

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Experiment no: 07

Name of the Experiment: Study of Operational Amplifier as RC Phase Shift Oscillator:

Objective:

A typical R-C phase shift oscillator will be constructed and studied in this experiment.

Equipment:

- IC (Op-Amp) 741 (1 Piece)
- Resistor → 1.5 k Ω (3 Pieces), 47k Ω (1 Piece)
- Capacitor → 0.01 μ F (3 Pieces)
- AT-7000 Trainer Board (1 Unit)
- DC Power Supply (1 Unit) → +15 volts, -15 Volts
- Oscilloscope (1 Unit)
- Multimeter (1 Unit)
- Chords and Wires

Theoretical Background:

The RC Phase Shift Oscillator is a sine wave generator that employs an amplifier in combination with a resistive-capacitive feedback network to produce sustained oscillations. In its most common form, three identical RC sections are cascaded to provide a total phase shift of 180°, which, combined with the 180° phase inversion of a common inverting amplifier stage, yields the 360° (or 0°) net phase shift required for oscillation. Each RC section contributes 60° of phase shift at the oscillation frequency, and the values of the resistors and capacitors in the network determine the frequency:

$$f = 1/2\pi RC\sqrt{6}$$

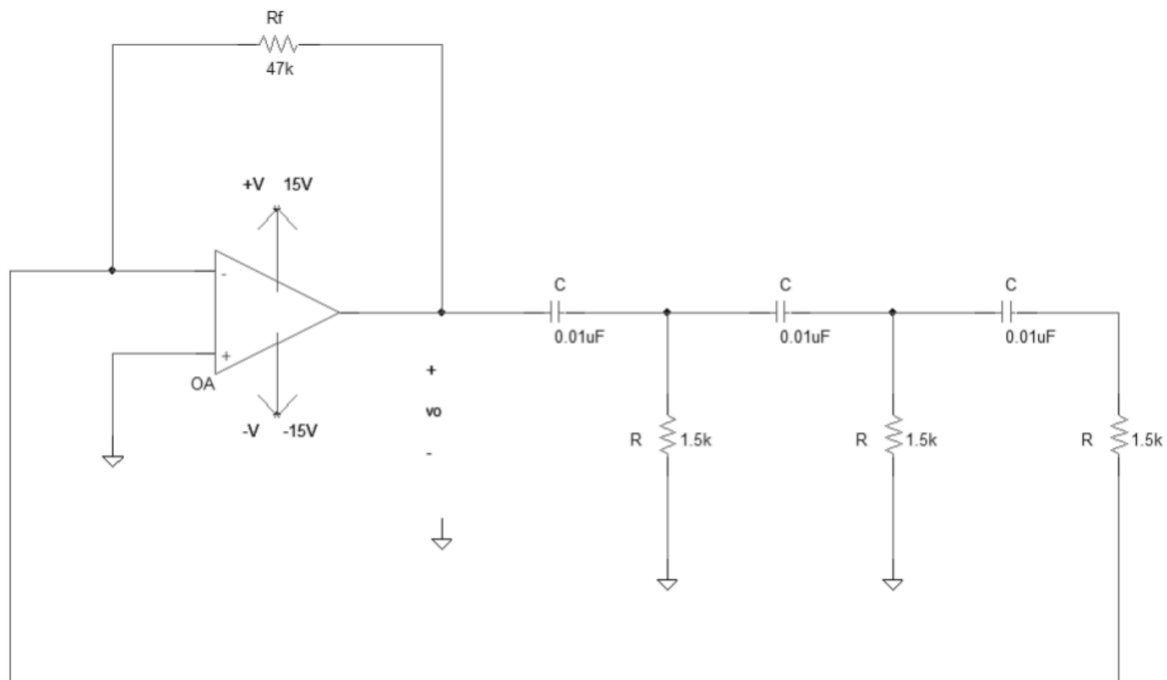
Here, R is the resistance in each section and C is the capacitance.

In this experiment, an op amp is configured as the inverting gain element, with feedback taken from the output through the RC ladder to the input. The op amp provides the necessary loop gain needed to compensate for losses in the RC network. The feedback resistor (R) controls this gain.

When powered and correctly biased, the network regenerates its own signal without an external input, producing a near sinusoidal output whose purity depends on the linearity of the amplifier and stability of component values.

RC phase-shift oscillators are favored for low to mid audio frequency applications due to their simplicity, ability to operate without inductors, and ease of tuning via resistor or capacitor variation. They find use in audio signal generation, instrumentation, and as reference sources where moderate frequency stability is acceptable.

Circuit Diagram:



Results and calculations:

$$C_1 = 0.011 \mu\text{F}$$

$$C_2 = 0.010 \mu\text{F}$$

$$C_3 = 0.010 \mu\text{F}$$

$$\begin{aligned}\therefore C_{\text{avg}} (\text{Determined for ease of calculation}) &= (C_1 + C_2 + C_3)/3 \\ &= (0.011 + 0.010 + 0.010)/3 \\ &= 0.0103 \mu\text{F}\end{aligned}$$

Measured values of the resistors (with multimeter):

$$R_1 = 1.49 \text{ k}\Omega$$

$$R_2 = 1.47 \text{ k}\Omega$$

$$R_3 = 1.49 \text{ k}\Omega$$

$$\begin{aligned}\therefore R_{\text{avg}} (\text{Determined for ease of calculation}) &= (R_1 + R_2 + R_3)/3 \\ &= (1.49 + 1.47 + 1.49)/3 \\ &= 1.483 \text{ k}\Omega\end{aligned}$$

Now, calculating the frequency, we get,

$$f = \frac{1}{2\pi RC\sqrt{6}} = \frac{1}{2\pi \times R_{\text{avg}} \times C_{\text{avg}} \times \sqrt{6}} = \frac{1}{2\pi \times 1.483 \times 10^3 \times 0.0103 \times 10^{-6} \times \sqrt{6}} = 4258.63 \text{ Hz}$$

From Oscilloscope :

Obtained Frequency=4.275kHz

Peak to Peak Voltage=27.6V

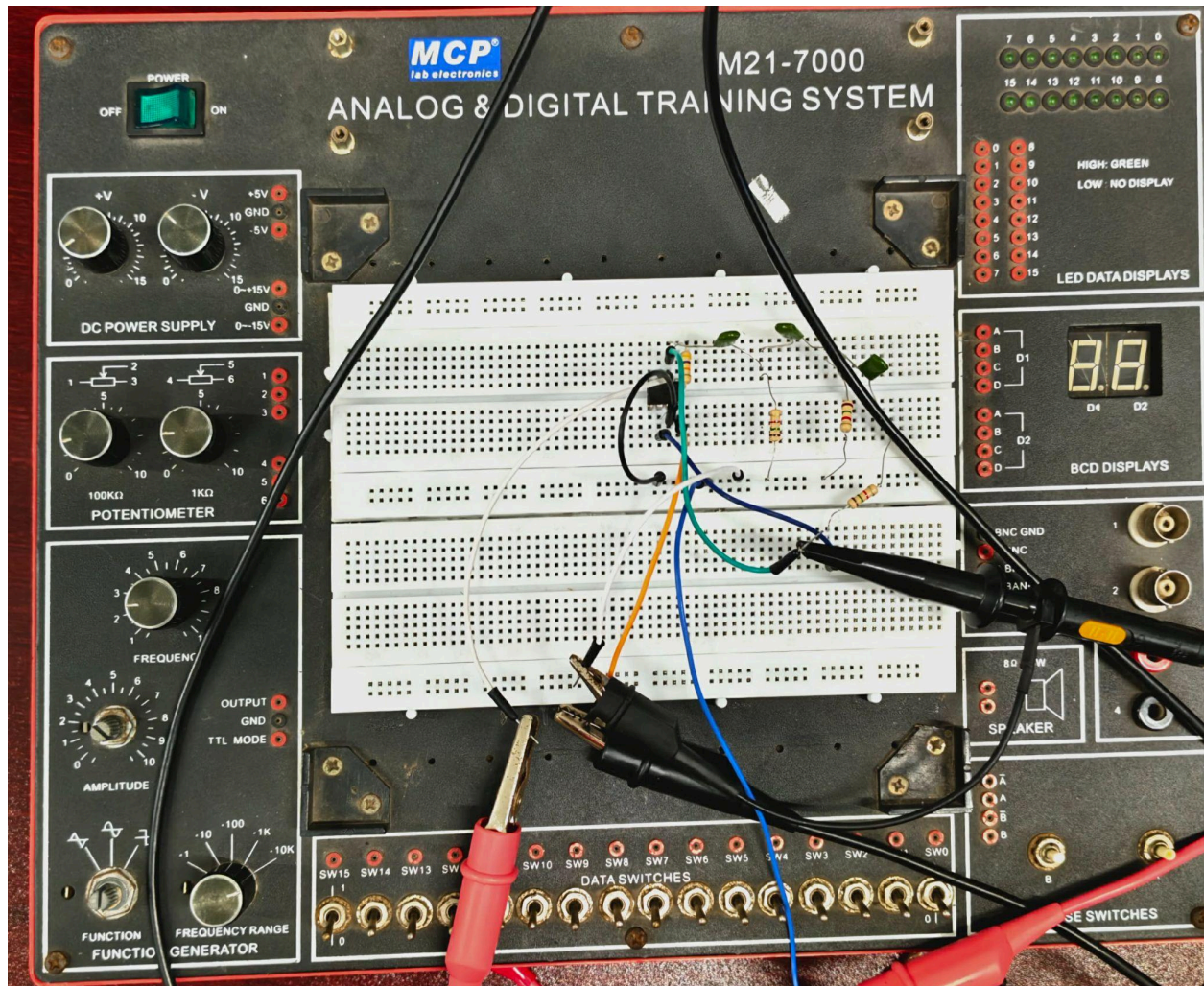


Fig-1: Hardware Circuit Diagram

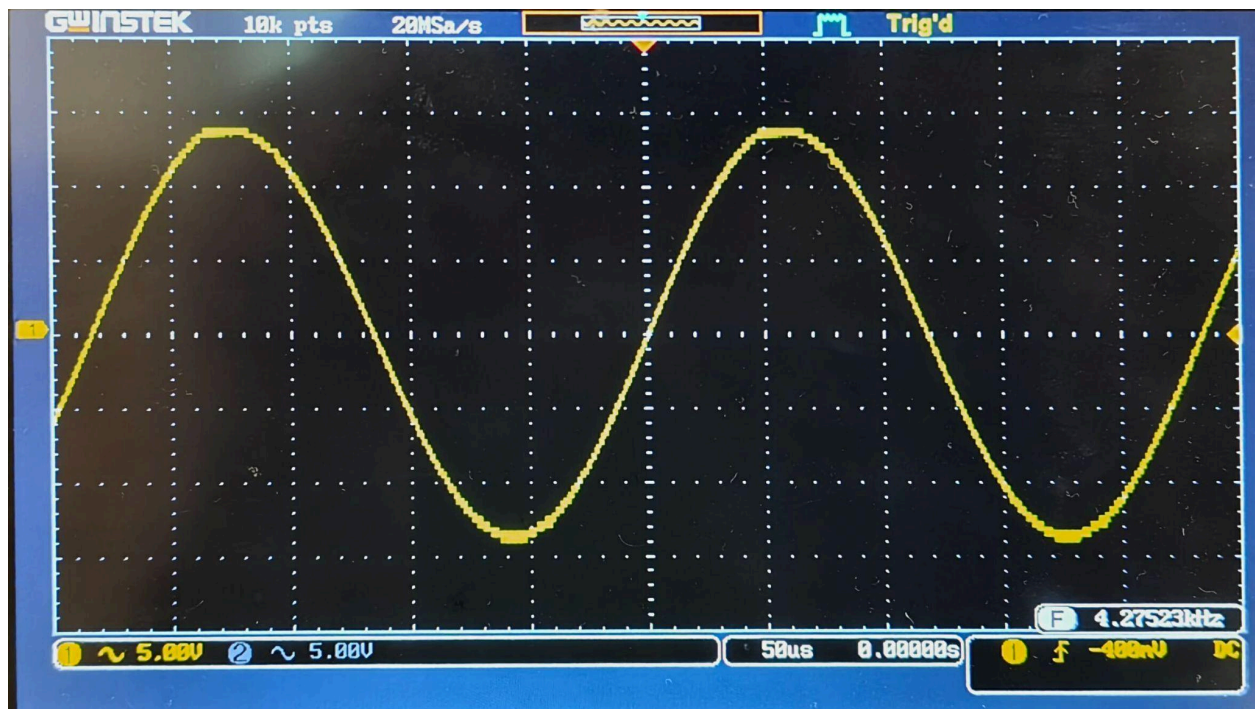


Fig-2: Output waveform

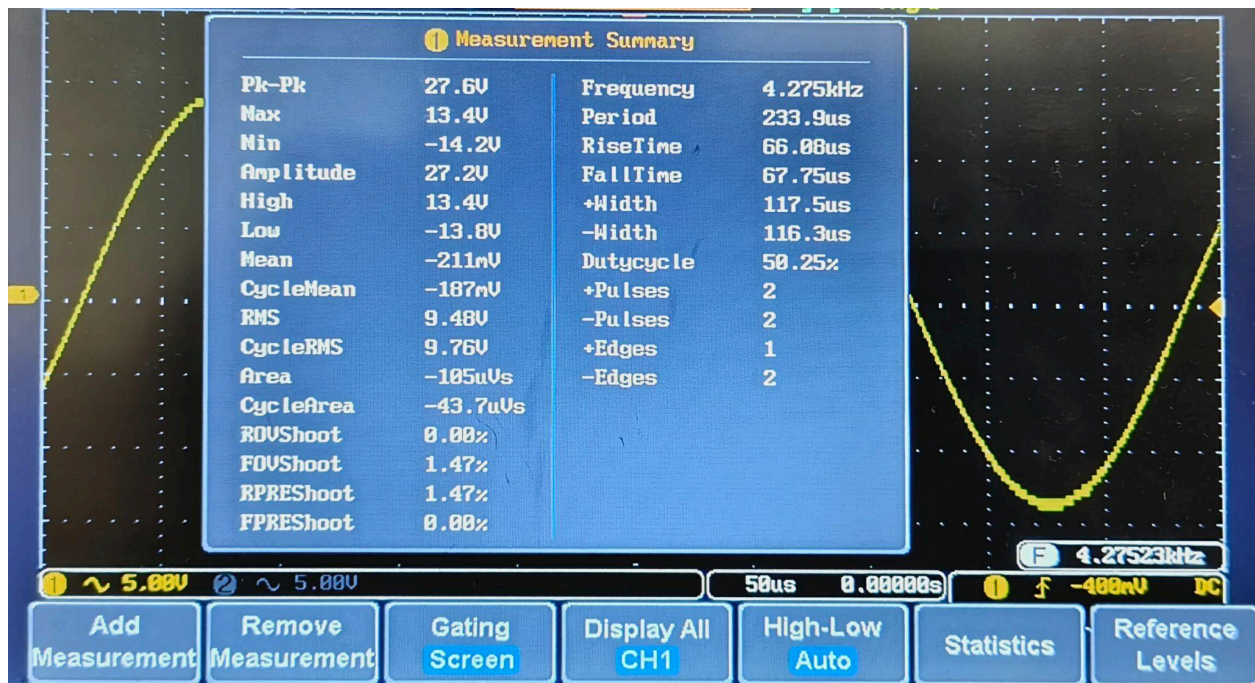


Fig-3: Measurement of the Output Waveform

Discussion:

The experiment successfully demonstrated the generation of a sinusoidal waveform using an op-amp and a three-stage RC feedback network. Upon constructing the circuit with equal resistor and capacitor values, the output waveform was observed to be stable and periodic, confirming the oscillator's ability to sustain oscillations without external input. The measured frequency is quite a bit different compared to that of the theoretical value calculated using the standard formula, which is due to the difference in the values of the capacitors and resistors. As we know, resistors and capacitors (most of the time) don't have the rated value, there are slight deviations from the rated one with the actual one.

Minor deviations in amplitude and waveform symmetry were noted, likely due to component mismatches and op amp limitations. The phase shift across each RC stage was approximately 60° , as expected, enabling the required 180° feedback shift. Overall, the experiment reinforced the practical application of phase shift principles in oscillator design and highlighted the importance of precise component selection for waveform fidelity and frequency stability.

Signed Data Sheet:

Group-4
Experiment-7

